
Building Legal Reward Models for Grounding and Abstention

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Abstract

Large language models are increasingly used in high-stakes domains such as law, where systems must **ground their reasoning in evidence** and **abstain under uncertainty**. However, existing reward models prioritise helpfulness and fluency rather than contextual grounding, often producing unsupported legal reasoning in retrieval-augmented generation (RAG) settings. We introduce LEGAL-REWARD BENCH (LRB), a legal contextual reward modelling benchmark constructed from legal QA datasets for evaluating **grounded legal generation under noisy and insufficient retrieval conditions**. Using 15 open-weight reward models (0.5B–9B), we study contextual reward modelling for grounded legal evaluation using moderate scale open-source models. Experiments on both CONTEXTUAL JUDGE BENCH (CJB) and LRB show that contextual DPO refinement improves grounded legal evaluation by up to +25.6 pp over baseline reward models. Performance is sensitive to dataset construction, with both length-balanced augmentation and combining general contextual with legal preference data improving evaluation accuracy. Models trained primarily on Victorian criminal law also transfer to external US legal benchmarks, improving HOUSING STATUTE QA by +16.2 pp. Our results suggest that grounded contextual reasoning behaviours can transfer across legal domains without frontier-scale systems, providing a reproducible foundation for evaluating grounded legal reasoning in retrieval-augmented settings.

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1. Introduction

Large language models (LLMs) are increasingly deployed in high-stakes domains such as law, medicine, and finance, where generated outputs may influence legal decisions, clinical care, and access to essential services (Siino et al., 2025; Costa-Gomes et al., 2026; Chen et al., 2024). Unlike open-domain conversational settings, these domains require reasoning grounded in external evidence and constrained by domain-specific rules. In such settings, useful systems must not only produce correct answers, but also remain faithful to the provided context and refuse to answer when the available evidence is insufficient (Pipitone & Alami, 2024; Peng et al., 2025). However, current language models are **primarily optimised for fluent and helpful interaction, often over-relying on parametric knowledge and failing to abstain under uncertainty**.

Key argument

Retrieval-augmented generation (RAG) partially addresses these limitations by providing models with access to external documents and domain-specific evidence (Lewis et al., 2020). However, retrieval alone **does not guarantee grounded reasoning**. Models may **ignore retrieved evidence, rely on memorised parametric knowledge, or generate plausible but unsupported conclusions** that are not justified by the provided context (Zheng et al., 2025; Huang et al., 2025). The problem becomes particularly severe when retrieved documents are ambiguous, incomplete, or noisy, as models often continue answering rather than abstaining under uncertainty (Feng et al., 2024; Peng et al., 2025; Butler & Butler, 2026). Grounded generation therefore requires mechanisms that evaluate whether responses faithfully use retrieved evidence.

Legal reasoning further intensifies the challenges of grounded generation. Legal systems are jurisdiction-dependent, procedurally constrained, and require reasoning over precedents, statutes, and factual scenarios (Zhong et al., 2020; Katz et al., 2023; Guha et al., 2023). Answers rarely reduce to purely deductive reasoning, instead requiring contextual interpretation and the weighing of competing legal considerations (Guha et al., 2023; Katz et al., 2024;

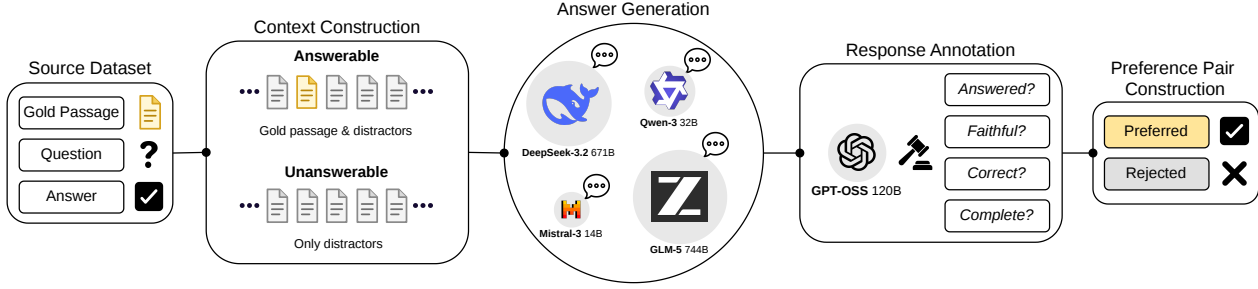


Figure 1. Legal preference-pair construction pipeline. Starting from Legal RAG Bench triples (c_{gold}, q, a) (Section 2.1), we construct six retrieved-context variants by combining the gold passage with distractor passages to control answerability and retrieval noise (Section 2.2). Four LLMs generate candidate responses across all (c, q) instances (Section 2.3). A judge-model assigns pointwise labels for answer behaviour, faithfulness, correctness, and completeness (Section 2.4). A hierarchical preference rule then converts labelled responses into contextual preference pairs (c, q, r^+, r^-) for reward-model training and evaluation (Section 2.5).

Pipitone & Alami, 2024; Fan et al., 2026). As a result, unsupported or hallucinated conclusions can appear highly convincing despite lacking justification in the underlying legal sources (Dahl et al., 2024). In such settings, **refusal is often preferable to generating confident but unsupported legal reasoning**.

Reinforcement learning from human feedback (RLHF) and reward modelling have become central approaches for aligning language models with human preferences (Ouyang et al., 2022; Rafailov et al., 2023; Bai et al., 2022). However, most existing reward models are trained on general-purpose preference data that prioritises helpfulness, fluency, and conversational quality rather than contextual grounding (Wang et al., 2024b; Lambert et al., 2025). As a result, **they often fail to reliably evaluate whether responses are supported by retrieved evidence or whether a model should abstain when context is insufficient** (Xu et al., 2025). Recent work on contextual reward modelling instead evaluates responses relative to retrieved evidence, substantially improving evaluation in retrieval-augmented settings (Xu et al., 2025; Coman et al., 2025). However, **reliable reward models and datasets for grounded legal reasoning remain limited**.

To address these challenges, we propose a **reproducible framework for contextual reward modelling in legal reasoning**. We construct a legal reward modelling dataset from existing legal question answering benchmarks by converting $(\text{context}, \text{question}, \text{answer})$ triples into structured preference data capturing answerability, groundedness, completeness, and legal correctness. Rather than treating legal reward modelling as simple domain adaptation, we frame it as the problem of **learning contextual grounding under retrieval uncertainty**: models must distinguish when retrieved evidence supports an answer, when it is incomplete or misleading, and when refusal is the preferred behaviour. Across both general and legal retrieval-augmented evaluation settings, we show that contextual reward modelling improves grounded evaluation and refusal behaviour, while also revealing that

benchmark construction strongly determines whether these behaviours are learnable and measurable. We make the following **contributions**:

- We introduce a **pipeline for constructing legal contextual reward modelling datasets** from legal QA benchmarks (Section 2). We release **LEGALREWARD-BENCH** (LRB), a 1,220-pair benchmark capturing refusal, faithfulness, completeness, and legal correctness in retrieval-augmented legal generation settings.
- We benchmark 15 open-weight reward models (0.5B–9B) on both CONTEXTUALJUDGE BENCH (CJB) (Section 4.1) and **LEGALREWARD BENCH** (Section 4.2), and show that contextual **DPO refinement substantially improves grounded legal evaluation**. Our experiments reveal that legal reward modelling is **sensitive to dataset construction**: both length-balanced augmentation and combining general contextual with legal preference data improve grounded evaluation, yielding gains of up to **+25.6 pp**.
- We demonstrate cross-jurisdiction transfer (Section 4.3) on external US legal benchmarks, where models trained primarily on Victorian criminal law improve performance on HOUSING STATUTE QA by **+16.2 pp**, suggesting that grounded contextual reasoning may transfer beyond a single legal domain.

2. Methodology

This section describes how we construct **LEGALREWARD-BENCH**, a contextual preference corpus for legal reward models. Starting from a legal QA dataset of (c_{gold}, q, a) triples consisting of a supporting passage, a question, and a reference answer (Section 2.1), we produce preference pairs (c, q, r^+, r^-) where c is a constructed retrieved context and r^+ is more strongly grounded in c than r^- . When c lacks the evidence to answer q , r^+ is an abstention rather than a

Failure Type	Question q	Preferred Response r^+	Rejected Response r^-
Refusal	What sentence was imposed on the accused?	“The provided context is insufficient to answer this question.”	“The accused was sentenced to five years imprisonment.”
Faithfulness	Which Act introduced cyberstalking offences?	“The Crimes (Stalking) Act 2003 introduced cyberstalking offences.”	“The Crimes Amendment Act 2001 introduced cyberstalking offences.”
Correctness	Does Victoria protect against double jeopardy?	“Yes. Victoria protects against double jeopardy.”	“No. Victoria does not recognise double jeopardy protections.”
Completeness	Must jurors be excused after seeing pre-trial publicity?	“No. Judges can usually address prejudice through directions to the jury.”	“No.”

Table 1. Legal preference pairs. Representative pairwise preference examples. In the dataset, r^+ and r^- are selected from generated candidate responses after pointwise annotation; they are not manually written reference or corrupted responses. Refusal examples test whether a model abstains when the retrieved evidence is insufficient. Faithfulness, correctness, and completeness examples reflect distinct failures of grounded reasoning. Full examples including retrieved context are provided in Appendix A.4.

fabricated answer.

The pipeline has four stages: context construction (Section 2.2), answer generation (Section 2.3), response annotation (Section 2.4), and preference pair construction (Section 2.5). For each source example we vary whether the retrieved context contains the supporting evidence, generate candidate responses with open-weight LLMs, label each response with a judge model along four contextual quality dimensions, and apply a hierarchical rule to select r^+ and r^- .

2.1. Source Dataset

We use Legal RAG Bench (Butler & Butler, 2026) as the substrate for dataset construction. Legal RAG Bench is built from the Victorian Criminal Charge Book and contains 100 expert-written questions over a corpus of 4,876 passages. Each question is paired with a supporting passage and a long-form reference answer. The questions were designed to be lexically dissimilar from their supporting evidence, so the task tests semantic retrieval and grounded reasoning rather than keyword matching.

2.2. Context Construction

For each source triple, we construct two kinds of retrieved context. *Answerable* contexts contain the supporting passage alongside nine distractor passages, with the supporting passage placed at a deterministically chosen slot. *Unanswerable* contexts remove the supporting passage and its topic family, replacing them with ten distractors. A passage’s *topic family* consists of passages sharing its leading two-component section number in the Charge Book (e.g. sections 5.7.3 and 5.7.11 share topic family 5.7), and is excluded from distractor sampling in both cases so that distractors cannot reintroduce the supporting evidence indirectly.

We vary the distractor source across three methods: uni-

form random sampling, BM25, and Nomic semantic retrieval (via `nomic-embed-text`). Answerable variants test grounded answering under retrieval noise, and unanswerable variants test abstention under insufficient evidence. This yields six variants (three answerable and three unanswerable), which form the main LegalRewardBench preference corpus. A distractor ablation extends the set with additional retrieval methods and gold-only baselines (Appendix A.3).

2.3. Answer Generation

We generate candidate responses by prompting four open-weight instruction-tuned models ranging from 14B to 744B parameters¹ to answer each question using only the retrieved context. Generation uses deterministic decoding with temperature 0.0 and a maximum length of 8,192 tokens, producing four candidate responses per (question, variant) cell, for a total of 2,400 candidate responses across 100 questions and six retrieval variants.

Rather than constructing r^- by manually corrupting the reference answer, we let failure modes emerge from the model panel: a candidate may abstain, answer correctly, omit material qualifications, introduce unsupported claims, or reach a conclusion contradicted by the retrieved passages. The reference answer is used only for downstream correctness and completeness labelling, not as a candidate. The generation prompt is in Appendix F.

2.4. Response Annotation

We label each candidate along four dimensions adapted from ContextualJudgeBench’s contextual evaluation framework (Xu et al., 2025): answer behaviour, faithfulness, correctness, and completeness. We add correctness because a

¹Qwen 3 32B, DeepSeek V3.2, GLM-5, and Ministral 3 14B, served via DigitalOcean serverless.

response can be grounded in the retrieved context yet still reach the wrong legal conclusion, and we drop conciseness because it is less tied to grounding and often subjective in legal QA.

Labelling is performed by GPT-OSS-120B using structured prompts. The judge decodes deterministically (temperature 0.0, 2,048-token cap), and each label is produced by a separate LLM call so that dimensions are scored independently. These labels are intended to support scalable contextual preference construction rather than replace expert legal assessment. Annotation proceeds in two stages.

We first assign an answer-behaviour label, capturing what the model attempted to do independent of correctness: *attempted* (the response tries to answer), *abstained* (the response declines, citing insufficient or unavailable evidence), or *unusable* (empty, unreadable, incoherent, malformed, or severely repetitive output).

Only attempted responses receive semantic labels:

- **Faithfulness** (fully supported, partially supported, unsupported, or contradicted): whether the response is supported by the retrieved context. The main conclusion governs the label: an unsupported or mischaracterised main conclusion is labelled unsupported, while an unsupported secondary claim is labelled partially supported.
- **Correctness** (correct or incorrect): whether the bottom-line conclusion matches the reference answer.
- **Completeness** (complete or incomplete): whether the response includes the essential rule, condition, exception, or qualification needed to justify the conclusion.

Abstained and unusable responses receive no semantic score, since they contain no claims to evaluate. The annotation prompts are in Appendix F.

2.5. Preference Pair Construction

Within each (context, question) cell, we compare every unordered pair of labelled candidate responses under a hierarchical preference rule. When the rule yields a strict preference, the higher-ranked response becomes r^+ and the lower-ranked becomes r^- . Ties are discarded. The hierarchy depends on context answerability. For answerable contexts, responses are ranked lexicographically by answer behaviour, then faithfulness, then correctness, then completeness. The preferred values are attempted, fully supported, correct, and complete respectively. For unanswerable contexts, only answer behaviour is used, and the preference reverses: abstained becomes the preferred value. Applied to 100 questions and six retrieval variants, this yields 3,600 unordered comparisons. 1,220 yield a strict

preference, and the rest are ties. We bucket each pair by the first hierarchy dimension on which the chosen and rejected responses differ. The per-bucket distribution is reported in Appendix 4.

3. Experimental Setup

Our experiments evaluate grounded legal generation under adversarial retrieval conditions. Rather than benchmarking retrieval quality itself, we study whether models in retrieval-augmented generation (RAG) settings can appropriately abstain when evidence is insufficient, remain faithful when relevant evidence is mixed with distracting context, and avoid unsupported legal conclusions under noisy retrieval conditions.

We evaluate reward models on two contextual preference benchmarks. First, we use CONTEXTUALJUDGE BENCH (CJB) (Xu et al., 2025), a general-domain benchmark for evaluating reward models in retrieval-augmented generation settings. CJB contains 2,000 preference pairs, with 1,600 training pairs and 400 test pairs ($n = 373$ retained after excluding examples exceeding the 4,096-token evaluation limit) covering refusal, faithfulness, completeness, and conciseness under both answerable and unanswerable retrieval conditions.

Second, we introduce LEGALREWARD BENCH (LRB), a legal-domain contextual reward modelling benchmark constructed from legal question answering datasets. LRB contains 1,220 preference pairs split into 940 training, 124 development, and 156 test examples, spanning both general grounding failures and domain-specific Victorian criminal law reasoning. Each example consists of a tuple (c, q, r^+, r^-) , where the rejected response r^- captures failures such as incorrect refusal, hallucination, incompleteness, or unsupported legal conclusions. Full dataset statistics and split details are provided in Appendix A.1.

Reward Model Scoring Let $\mathbf{p} = (p, c, q)$ denote a prompt, retrieved context, and question, and let $r = (r_1, \dots, r_T)$ denote a candidate response of length T . A reward model parameterised by θ assigns a scalar score

$$s_\theta(\mathbf{p}, r) \in \mathbb{R}, \quad (1)$$

where higher values indicate stronger preference for the response. For causal language models, reward scores are commonly computed using the unnormalised sequence log-probability, which sums token log-probabilities across the response. This matches the standard formulation used in DPO and prior reward-model benchmarks (Rafailov et al., 2023; Lambert et al., 2025; Xu et al., 2025). However, because summed log-probabilities scale with response length, this formulation can systematically favour shorter responses

when preference pairs exhibit substantial length disparities. We therefore use two complementary scoring functions depending on the benchmark construction.

$$s_{\theta}(\mathbf{p}, r) = \sum_{t=1}^T \log p_{\theta}(r_t \mid \mathbf{p}, r_{<t}), \quad (2)$$

where $r_{<t} = (r_1, \dots, r_{t-1})$, matching the convention of prior reward-model benchmarks and the DPO training objective (Rafailov et al., 2023; Lambert et al., 2025; Xu et al., 2025). On LEGALREWARDBENCH (Section 2), where residual length variation in augmented refusal templates would otherwise dominate sum-based comparisons, we use the length-normalised log-probability

$$\ell_{\theta}(\mathbf{p}, r) = \frac{1}{T} \sum_{t=1}^T \log p_{\theta}(r_t \mid \mathbf{p}, r_{<t}). \quad (3)$$

Specialist reward models with a scalar reward head use the head output directly in place of s_{θ} . Given a preference pair (r^+, r^-) , a prediction is considered correct when

$$s_{\theta}(\mathbf{p}, r^+) > s_{\theta}(\mathbf{p}, r^-). \quad (4)$$

For an contextual preference pair dataset (c_i, q_i, r_i^+, r_i^-) where $i \in \{1, \dots, N\}$ and prompt p , we report pairwise accuracy

$$\text{Acc} = \frac{1}{N} \sum_{i=1}^N \mathbf{1}[s_{\theta}(\mathbf{p}_i, r_i^+) > s_{\theta}(\mathbf{p}_i, r_i^-)], \quad (5)$$

where $\mathbf{1}[\cdot]$ denotes the indicator function. Random performance corresponds to 50% accuracy. We report Wilson 95% confidence intervals and overall pairwise accuracy; macro-average across splits is computed in evaluation outputs but omitted from tables for conciseness.

Training Objective We fine-tune models using Direct Preference Optimisation (DPO) (Rafailov et al., 2023), which directly optimises the relative preference margin between preferred r^+ and rejected r^- responses conditioned on the same prompt, context and question. Because both training and evaluation rely on the same underlying response scoring function, pairwise accuracy directly measures whether the model assigns higher reward to preferred grounded legal responses. We additionally explored with supervised fine-tuning (SFT) on preference pairs, but found it consistently less effective than DPO (Appendix B.3).

Models We evaluate 14 open-weight instruction-tuned models spanning the 0.5B–27B parameter range, including models from the Qwen (Qwen et al., 2025; Qwen Team,



Figure 2. DPO refinement improves grounded evaluation on CONTEXTUALJUDGE BENCH. Compared to the Ministral-8B baseline (grey), contextual DPO refinement on CJB (gold) improves performance across the three contextual dimensions we focus on: refusal, faithfulness, and completeness.

2026), LLaMA (Meta, 2026), Gemma (Team et al., 2024; 2025), Phi (Abdin et al., 2024; Microsoft et al., 2025), Ministral (Liu et al., 2026), DeepSeek-R1-distilled (DeepSeek-AI et al., 2025), SFR-RM-R (Dong et al., 2024), and SmolLM families (Allal et al., 2025). We additionally compare against specialist reward models designed for preference evaluation and LLM-as-a-judge tasks, including Skywork-Reward (Liu et al., 2024), Selene (Alexandru et al., 2025), and SFR-Judge (Wang et al., 2024a). For finetuning for contextual reward modelling, we fine-tune a representative subset of models spanning different architectures and scales: Phi4-mini (Microsoft et al., 2025), DeepSeek-R1-8B (DeepSeek-AI et al., 2025), Qwen3.5-9B (Qwen Team, 2026), Qwen3.5-27B (Qwen Team, 2026), and Ministral-8B (Liu et al., 2026). All models were evaluated under identical hardware conditions. Full implementation and compute details are provided in Appendix D.1.

4. Results

4.1. Contextual Evaluation

We first evaluate 14 open-weight instruction-tuned models together with specialist reward models on CONTEXTUAL-JUDGE BENCH (CJB). Full per-dimension results are provided in Appendix B.1. Across models, overall performance remains only marginally above random chance, with no single baseline consistently outperforming the others across all grounding dimensions. Most models exhibit a strong pref-

Table 2. Legal contextual reward modelling is sensitive to dataset construction. Evaluation on LEGALREWARDBENCH LRB. Length-balanced augmentation and general grounded pairs improves grounded legal evaluation, particularly for baseline Ministral-8B. Higher is better. Wilson 95% confidence intervals are provided in Appendix - Table 10. Best results are shown in **bold**; second-best results are underlined.

Training	Gemma 2-2B		Ministral-8B	
	Overall	Δ	Overall	Δ
Random	50.0	–	50.0	–
Baseline	61.8	–	59.3	–
CJB	66.4	+4.6	67.6	+8.3
LRB	63.6	+1.8	61.9	+2.6
CJB+LRB	65.7	+3.9	66.0	+6.7
LRB-Length-Aug	81.1	+19.3	81.8	+22.5
CJB+LRB-Length-Aug	81.4	+19.6	84.9	+25.6

erence for shorter responses, resulting in high conciseness accuracy but substantially weaker completeness and refusal behaviour. Based on these baseline experiments, we select a representative subset of models for contextual DPO refinement. We additionally explored supervised fine-tuning (Appendix B.3) but found it consistently less effective than DPO. Among the evaluated baselines, Ministral-8B achieved the strongest balance across faithfulness, completeness, and refusal despite its relatively modest parameter scale, consistent with observations reported by RAGfreee (Coman et al., 2025). Figure 2 shows that **DPO refinement improves grounded evaluation performance across contextual dimensions**, particularly refusal behaviour under insufficient retrieval conditions.

4.2. Legal Contextual Reward Modelling

We next evaluate contextual reward modelling in the legal domain using LEGALREWARDBENCH (LRB). Compared to CONTEXTUALJUDGE BENCH (CJB), legal contextual evaluation is more challenging due to smaller dataset scale, domain-specific reasoning requirements, and stronger structural biases in refusal-style responses. Table 2 shows that DPO refinement on CJB transfers to legal contextual evaluation, improving both Gemma 2-2B and Ministral-8B despite never observing legal-domain preference pairs during optimisation. Interestingly, CJB-only refinement outperforms legal-only DPO training, **suggesting that grounded contextual evaluation and refusal behaviour are partially domain-general capabilities that transfer across retrieval-augmented settings**.

Simply combining CJB with the original LRB pairs does not consistently improve performance, despite increasing the amount of preference data. This suggests that the limiting factor is not only data volume, but also the structure of the legal preference pairs. Inspection of LRB revealed



Figure 3. Length-balanced augmentation improves legal contextual reward modelling. Compared to the Ministral-8B baseline (grey), DPO refinement on CJB alone transfers strongly to LEGALREWARDBENCH (blue). Joint training on CJB and LRB (pink) is limited by structural response-length artefacts in the legal preference data. After applying length-balanced augmentation (gold), performance improves substantially across evaluation dimensions.

systematic response-length asymmetries, especially in refusal cases, where abstentions and substantive legal answers differ sharply in length. Under sequence log-probability scoring, models can exploit these superficial length cues rather than learning contextual grounding and refusal behaviour. **Legal reward modelling is sensitive to dataset construction and preference-pair structure.**

To mitigate these effects, we introduce a length-balanced augmentation strategy (Appendix F) that rewrites refusal and negative responses using controlled templates designed to better match the length and structure of preferred grounded answers while preserving the underlying grounding failure. As shown in Figure 3 and Table 2, **length-balanced augmentation reduces response-length asymmetries and improves grounded legal evaluation.**

4.3. Cross-Jurisdiction Transfer

We evaluate whether contextual reward models trained primarily on Victorian criminal law generalise across jurisdictions and legal tasks. We evaluate transfer on two US legal reasoning benchmarks from the Stanford RegLab suite (Zheng et al., 2025): BAR EXAM QA, consisting of Multi-state Bar Examination questions paired with supporting US

Table 3. Contextual reward models partially generalise across legal jurisdictions. Models refined on CJB and LRB improve grounded evaluation on external US legal reasoning QA benchmarks despite training primarily on Victorian criminal law data. Results report pairwise reward accuracy with Wilson 95% confidence intervals. Higher is better. Best results are shown in **bold**; second-best results are underlined.

Model	Bar Exam	Δ	Housing	Δ
Random	50.0	–	50.0	–
Ministral-8B	56.4 [47,65]	–	54.2 [50,58]	–
+DPO	<u>60.7</u> [52,69]	+4.3	70.4 [66,74]	+16.2
Qwen3.5-9B	54.7 [46,63]	–	52.8 [48,57]	–
+DPO	54.7 [46,63]	+0.0	<u>55.4</u> [51,60]	+2.6
Qwen3.5-27B	60.7 [52,69]	–	50.8 [46,55]	–
+DPO	61.5 [52,70]	+0.8	53.0 [49,57]	+2.2

caselaw passages, and HOUSING STATUTE QA, a benchmark of yes/no questions grounded in US state housing statutes. Both benchmarks differ substantially from the Victorian criminal law data used during training, providing an out-of-distribution test of contextual grounding and refusal behaviour.

Table 3 shows that **contextual reward models exhibit meaningful cross-jurisdiction transfer despite being trained primarily on Victorian criminal law data**. Gains are strongest on HOUSING STATUTE QA, where Ministral-8B improves by +16.2 *pp*, comparable to its in-domain gains on CONTEXTUALJUDGE BENCH. Improvements on BAR EXAM QA are smaller, suggesting that grounded contextual evaluation and refusal behaviour transfer more robustly than jurisdiction-specific legal reasoning itself. Interestingly, transfer does not scale uniformly with model size: refined Ministral-8B substantially outperforms larger Qwen models, indicating that contextual alignment is a stronger driver of grounded evaluation than parameter count alone.

5. Discussion

Our results suggest that grounded legal evaluation depends less on memorising jurisdiction-specific legal knowledge than on learning robust contextual reasoning and refusal behaviour. Across both general contextual evaluation (4.1) and legal reward modelling experiments (4.2), contextual preference optimisation substantially improves reliability under insufficient and noisy retrieval conditions. The cross-jurisdiction transfer results (4.3) further suggest that groundedness is at least partially a transferable behavioural capability rather than purely a function of model scale or domain memorisation. This distinction is important in legal settings, where reliable abstention is often preferable to generating confident but unsupported conclusions. More broadly, these findings suggest that trustworthy legal AI may be achievable

through targeted contextual alignment of relatively small open models.

Our experiments also show that legal reward modelling is highly sensitive to benchmark construction and evaluation methodology (4.2). In particular, response-length asymmetries in refusal datasets can distort pairwise reward evaluation under DPO sequence log-probability scoring, obscuring improvements in grounded behaviour. Length-balanced augmentation (F) substantially mitigates this effect, suggesting that some limitations of legal reward models may reflect weaknesses in evaluation design rather than an absence of learnable signal. Future progress in grounded legal evaluation will therefore depend not only on improved models, but also on reward objectives and benchmarks that better separate groundedness, completeness, and verbosity.

6. Related Work

Legal Reasoning and Legal Language Models Recent advances in large language models have increased interest in legal AI (Devlin et al., 2019; OpenAI et al., 2024). Legal reasoning remains challenging due to jurisdiction-specific interpretation, long-context reasoning, and the requirement that conclusions remain grounded in authoritative sources (Katz et al., 2023; Zhong et al., 2020). Legal NLP has evolved from classification and judgement prediction toward large-scale reasoning benchmarks such as LexGLUE, LegalBench, LEXam, and PLAWBench (Chalkidis et al., 2022; Guha et al., 2023; Fan et al., 2026; Shi et al., 2026). However, recent evaluations of frontier systems continue to reveal hallucinated authorities, unsupported conclusions, and unreliable grounding despite strong general reasoning performance (Dahl et al., 2024; Katz et al., 2024).

Retrieval-Augmented Generation in Law Retrieval-augmented generation (RAG) enables language models to condition responses on retrieved evidence rather than parametric memory alone (Lewis et al., 2020). This paradigm is particularly important in law, where relevant authority is distributed across statutes, regulations, and case law. Recent work has introduced legal-specific retrieval and RAG benchmarks, including LegalBench-RAG and Legal RAG Bench (Pipitone & Alami, 2024; Butler & Butler, 2026). Although retrieval improves access to relevant legal evidence, current systems frequently remain unfaithful to retrieved context, hallucinate unsupported authorities, and fail to abstain under insufficient evidence (Butler & Butler, 2026; Peng et al., 2025).

Reward Modelling and LLM-as-a-Judge Reward modelling and RLHF are widely used to align language models with human preferences (Ouyang et al., 2022; Rafailov et al., 2023). Benchmarks such as RewardBench and JudgeBench

evaluate reward models across reasoning and factuality tasks (Lambert et al., 2025; Tan et al., 2024). However, most existing reward models remain largely non-contextual, prioritising helpfulness and stylistic preference rather than evaluating whether responses are grounded in retrieved evidence. Recent work on contextual evaluation begins to address this limitation. ContextualJudgeBench introduces evaluation for grounded generation under retrieval uncertainty, while RAGfree demonstrates that contextual reward models outperform general-purpose judges in RAG settings (Xu et al., 2025; Coman et al., 2025). Nevertheless, grounded reward modelling for legal reasoning remains comparatively underexplored, particularly for open and compute-efficient models.

7. Conclusion

We introduced LEGALREWARDBENCH, a reproducible framework for evaluating whether legal reward models ground responses in retrieved evidence under noisy and insufficient retrieval conditions (Section 2). Across general and legal retrieval-augmented benchmarks, contextual DPO refinement improved grounded evaluation and refusal behaviour, while revealing that performance is sensitive to dataset construction (Sections 4.1–4.2). In particular, response-length asymmetries can distort evaluation, whereas length-balanced augmentation and mixed contextual training improve performance. Cross-jurisdiction experiments further suggest that contextual grounding and refusal behaviour can transfer beyond Victorian criminal law, though not uniformly across tasks or model families (Section 4.3). Together, these findings provide a reproducible foundation for evaluating grounded legal reasoning in retrieval-augmented systems.

Limitations

Our work focuses on contextual reward modelling and grounded evaluation rather than deployment-ready legal reasoning systems. Although LEGALREWARDBENCH is constructed from legal QA datasets and grounded legal sources, we do not evaluate model preferences with practising lawyers or legal experts. Preference labels are derived through scalable benchmark construction rather than large-scale expert annotation, and therefore primarily measure groundedness, refusal behaviour, and consistency with reference answers rather than broader legal persuasiveness or professional legal judgement. Human evaluation with legal practitioners remains important future work.

Limitations also arise from the benchmark construction and reward modelling methodology itself. Sequence log-probability scoring structurally penalises longer responses, particularly affecting completeness evaluation where correct

legal answers are inherently longer and more variable than incomplete or refusal responses. Although length-balanced augmentation substantially mitigates this effect, the augmented benchmark modifies the original test distribution and should therefore be reported alongside the original evaluation rather than replacing it. In addition, Bar Exam QA and Housing Statute QA are converted into pairwise preference data, which approximates but does not fully reproduce open-ended legal generation settings.

Finally, our experiments remain limited in architectural diversity, scale, and jurisdictional coverage. While we observe meaningful transfer beyond Victorian criminal law, broader evaluation across additional legal systems, languages, and procedural settings remains necessary. Future work should explore explicit scalar reward heads, completeness-aware objectives, and richer expert-supervised evaluation that better disentangles groundedness, completeness, and legal reasoning quality.

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